

BCE-C612
SEMESTER EXAMINATION-MAY2023
CLASS: B.TECH SEMESTER: VI
FORMAL LANGUAGES AND AUTOMATA THEORY

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A

(SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: Define Non-Deterministic Finite Automata

Question-2: Define context free grammar

Question-3: Convert regular expression (01^*+1) to finite automata

Question-4: What is Chomsky Normal Form? Discuss.

Question-5: Bring out the differences of Melay and Moore Machine?

Question-6: Define Push Down Automata.

Question-7: Explain about types of Turing Machine.

Question-8: Discuss in detail about P and NP problems

Question-9: What is undecidable problem?

Question-10: Distinguish between class P and class NP Problems.

SECTION-B
(LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 X 10 = 40 Marks)

- Question-1: Design a DFA which accepts set of all strings which are divisible by 5 for binary number.
- Question-2: Explain about Pumping Lemma and write the Applications of the Pumping Lemma?
- Question-3: Define Regular Expression? Explain about the Properties of Regular Expressions.
- Question-4: Construct NFA with ϵ which accepts a language consisting the strings of any number of 0's followed by any number of 1's followed by any number of 2's And also convert into NFA without ϵ transitions.
- Question-5: Construct the regular expression corresponding to the language accepted by following DFA.



- Question-6: For the Grammar $\{S \rightarrow AS/a, A \rightarrow SbA/SS/ba\}$ construct Left most derivation for the string aabbbaaa?
- Question-7: Define PDA and construct PDA from the following CFG
 $S \rightarrow aAA$
 $A \rightarrow aS|bS|a$
- Question-8: Define Turing Machine and Construct a Turing Machine that will accept the Language consists of all palindromes of 0's and 1's?

Sessional Exam – I
Subject: Formal Languages and Automata Theory
Paper Code: BCE-C612

Time: 1 Hr

MM: 20

Answer any three questions from Section A (each question carries 4 marks).

Answer any one question from Section B (each question carries 8 marks).

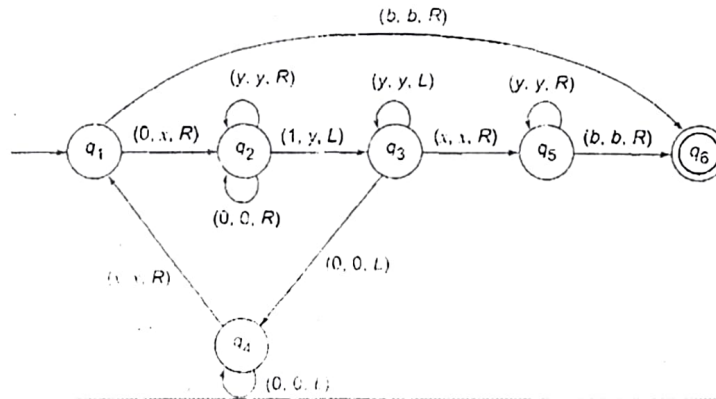
Q.No.	Section A (Questions of 6 marks each)	CO																								
Q1.	Identify the differences between DFA & NFA or Describe a Finite Automata with e-moves.	CO1																								
Q2.	Define Finite Automaton with the help of diagram. Or Describe the transition system	CO1																								
	Section B (Questions of 8 marks each)																									
Q3.	Compare the Mealy and Moore Machine and convert the following Moore machine to equivalent Mealy Machine <table border="1"><thead><tr><th>Present State</th><th colspan="2">Next State</th><th>Output</th></tr><tr><th></th><th>a=0</th><th>a=1</th><th></th></tr></thead><tbody><tr><td>->q0</td><td>q3</td><td>q1</td><td>0</td></tr><tr><td>q1</td><td>q1</td><td>q2</td><td>1</td></tr><tr><td>q2</td><td>q2</td><td>q3</td><td>0</td></tr><tr><td>q3</td><td>q3</td><td>q3</td><td>0</td></tr></tbody></table> Or Explain Grammer with its definition and If $G = (\{S\}, \{0,1\}, \{S \rightarrow 0S1, S \rightarrow \Lambda\}, S)$, find $L(G)$	Present State	Next State		Output		a=0	a=1		->q0	q3	q1	0	q1	q1	q2	1	q2	q2	q3	0	q3	q3	q3	0	CO2
Present State	Next State		Output																							
	a=0	a=1																								
->q0	q3	q1	0																							
q1	q1	q2	1																							
q2	q2	q3	0																							
q3	q3	q3	0																							

Second Sessional Exam
Subject: Formal Languages and Automata Theory
Paper Code: BCE-C612

Time: 1 Hr

MM: 20

Q.No.	Section A (Questions of 6 marks each)	CO
Q1.	What is a pushdown automata? Explain Or What is a Turing machine? Explain	CO3
Q2.	Construct a pda A accepting $L = \{wcw^T \mid w \in \{a,b\}^*\}$ by final state. Or Describe the different types of Turing Machines.	CO3
	Section B (Questions of 8 marks each)	
Q3.	Construct a pda A accepting the set of all strings over $\{a, b\}$ with equal number of a's and b's. or M is a Turing machine represented by the transition system in Fig. given below. Obtain the computation sequence of M for processing the input string 0011.	CO4



BCE-C612
SEMESTER EXAMINATION- MAY 2024
B. TECH SEMESTER: VI
FORMAL LANGUAGES AND AUTOMATA THEORY

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: A and B. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.

(5×6=30 Marks)

- Q-1 Define an Automaton with proper diagram. •
- Q-2 Illustrate the generalized transition graph with the help of suitable example.
- Q-3 Describe a Finite Automata with e-moves. •
- Q-4 Show whether $(0+1)^* 100$ is regular or not. •
- Q-5 Design a DFA to accept strings of a's and b's having a substring ab. •
- Q-6 Consider the following regular expression and construct the finite automaton
- $a + b$
 - $(a + b)^*$
 - $a(a + b)^*$
 - $a(a + b)^*b$
- Q-7 Differentiate PDA and non-deterministic PDA. •
- Q-8 Show that $\{a^n b^{2n} | n > 0\}$ is not a regular set, using pumping lemma.
- Q-9 List the two differences between DFA & NDFA. •
- Q-10 What are Universal Turing Machines? •

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4×10=40 Marks)

- Q-1 Explain Chomsky hierarchy in detail with suitable example

Q-2 Compare the Mealy and Moore Machine and convert the following Moore machine to equivalent Mealy Machine

Present State	Next State		Output
	a=0	a=1	
->q0	q3	q1	0
q1	q1	q2	1
q2	q2	q3	0
q3	q3	q3	0

Q-3 Define P, NP problems, NP complete and NP hard problems with example.

Q-4 Explain Grammar with its definition and

If $G = (\{S\}, \{0,1\}, \{S \rightarrow 0S1, S \rightarrow \Lambda\}, S)$, find $L(G)$

Q-5 State whether following languages are Regular Language or non-Regular Language with reasons:

$$L = \{a^n b^n : n \geq 0\}$$

Q-6 Design a DFA for the following language

$$L = \{0^m 1^n \mid m \geq 0 \text{ and } n \geq 1\}$$

Q-7 Construct the left-most and right-most derivations and parse trees for the following grammar

$$S \rightarrow aB \mid bA$$

$$A \rightarrow aS \mid bAA \mid a$$

$$B \rightarrow bS \mid aBB \mid b$$

which accepts the string "aaabbabbba".

Q-8 Discuss in detail (any two)

- Undecidability
- Types of Turing machines
- Greibach Normal form
- Context Free Grammars

Sessional 1 (B.Tech VI Sem CSE)
BCE-C612 Formal Languages and Automata Theory

Duration: 1HR

Date: 19/2/25

Time: 3:30 pm

Max. Marks :20

Section A

Note: Attempt any two questions. Each question carry 6 marks.

2 x 6 =12 M

Questions	LO	BT
Q1. Define the term Finite Automata. Differentiate between NFA and DFA.	CO1	U
Q2. Write regular expression for the language over the alphabet $\{0,1\}$, with atmost two consecutive zeros.	CO1	U
Q3. Given the following Finite State Automaton (FSA) $M=(Q,\Sigma,\delta,q_0,F)$ where: The transition function δ is defined as: $\delta(q_0,a)=q_1$, $\delta(q_0,b)=q_2$, $\delta(q_1,a)=q_0$, $\delta(q_1,b)=q_1$, $\delta(q_2,a)=q_2$, $\delta(q_2,b)=q_0$. The initial state and final state is q_0 . Is the string "ababb" accepted by the FSA.	CO3	E
Q4. Given the following Nondeterministic Finite Automaton (NFA) with null (ϵ) moves: Transition function δ $\delta(q_0,a)=\{q_0,q_1\}$, $\delta(q_0,b)=\{q_0\}$, $\delta(q_1,\epsilon)=\{q_2\}$, $\delta(q_2,b)=\{q_2\}$ $\delta(q_2,a)=\{q_1\}$. Initial state: q_0 , Accepting state: $F=\{q_2\}$ Remove the null (ϵ) transitions from the given NFA.	CO2	K

Section B

Note: Attempt any one question from this section.

1x8 = 8 M

Questions			LO	BT
Q1. Design a DFA equivalent for the given NFA; where q3 is the final state. [10]			CO1	U
Current State	Input 'a'	Input 'b'		
→ q0	q0, q1	q0 , q2		
q1	--	q3		
q2	q0, q3	q1		
q3	q2	-		
Q2. Given the following Nondeterministic Finite Automaton (NFA) $M=(Q,\Sigma,\delta,q_0,F)$.The transition function δ is defined as: $\delta(q_0,a)=\{q_0,q_1\}$, $\delta(q_0,b)=\{q_0\}$, $\delta(q_1,\epsilon)=\{q_2\}$, $\delta(q_2,b)=\{q_2\}$, $\delta(q_2,a)=\{q_1\}$.The initial state is q_0 .The set of accepting states is $F=\{q_2\}$ Is the string "ab" accepted by the NFA?			CO3	AP

Sessional II (B.Tech VI Sem CSE)

BCE-C612 Formal Languages and Automata Theory

Duration: 1HR

Date: 3/4/25

Time: 3:30 pm

Max. Marks :20

Section A

Note: Attempt any two questions. Each question carry 6 marks.

2 x 6 =12 M

Questions	LO	BT
Q1. Define the term Push Down Automata	CO1	U
Q2. Construct PDA which accepts strings made of 0 and 1 and is a pallindrome.	CO3	E
Q3.For the given grammar do left most derivation for the string aabaabbb ; $S \rightarrow ab \mid ba \mid aB \mid bA$; $A \rightarrow a \mid bAA$; $B \rightarrow b \mid aBB$.	CO2	k
Q4. Show the working of PDA by taking suitable example.	CO2	K

Section B

Note: Attempt any one question from this section.

1x8 = 8 M

Questions	LO	BT
Q1. Design CFG which accepts strings of type $a^i b^j$; $i=j$	CO3	E
Q2. Design a PDA for the string of type wcw^r , where w is made of a and b.	CO3	AP

PAPER CODE: BCE-C612
SEMESTER EXAMINATION (MAY-2025)
CLASS: B.TECH. SEMESTER: VI

PAPER TITLE: FORMAL LANGUAGES AND AUTOMATA THEORY

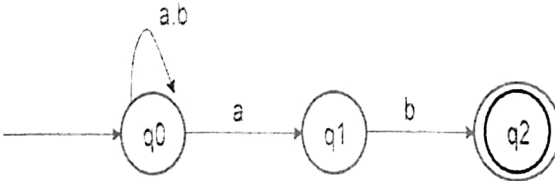
Time: 3 Hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 x 6 = 30 Marks)

		C O	B L
Q-1	Define finite automata. Differentiate between DFA and NFA.	1	U
Q-2	Define Push Down Automata.	1	U
Q-3	Write a regular expression for a string having no consecutive zeros. Made of 0 and 1.	2	
Q-4	Define Melay and Moore machines.	1	U
Q-5	Prove $10+(1010)^*[\wedge+(1010)^*]=10+(1010)^*$	3	K
Q-6	Draw a derivation tree for the input $id+id*id$ for the following grammar: $E \rightarrow E+T/T$ $T \rightarrow T*F/F$ $F \rightarrow (E)/id$	3	K
Q-7	Convert the following grammar into Greibach normal form. $S \rightarrow AA/a$ $A \rightarrow SS/b$	4	A P
Q-8	Convert the following NFA to DFA: 	4	A P
Q-9	For the NFA in question no 08, find whether $aaabaab$ is accepted or not.	1	U
Q-10	Explain 4 types of grammar.	1	U

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 x 10 = Marks)

Q-1	Consider the below finite automata and check the strings are accepted or not	3	A P																	
	<table border="1"> <thead> <tr> <th rowspan="2">States (Q)</th> <th colspan="2">Input Alphabtes</th> </tr> <tr> <th>0</th> <th>1</th> </tr> </thead> <tbody> <tr> <td>→q0</td> <td>q1</td> <td>q3</td> </tr> <tr> <td>q1</td> <td>q0</td> <td>q2</td> </tr> <tr> <td>q2</td> <td>q3</td> <td>q1</td> </tr> <tr> <td>q3</td> <td>q2</td> <td>q0</td> </tr> </tbody> </table> (i) 1110 (ii) 0001 (iii) 1010	States (Q)	Input Alphabtes		0	1	→q0	q1	q3	q1	q0	q2	q2	q3	q1	q3	q2	q0		
States (Q)	Input Alphabtes																			
	0	1																		
→q0	q1	q3																		
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q3	q2	q0																		
Q-2	Design a DFA equivalent for the given NFA; where q3 is the final state. <table border="1"> <thead> <tr> <th>Current State</th> <th>Input 'a'</th> <th>Input 'b'</th> </tr> </thead> <tbody> <tr> <td>→ q0</td> <td>q0, q1</td> <td>q0, q2</td> </tr> <tr> <td>q1</td> <td>--</td> <td>q3</td> </tr> <tr> <td>q2</td> <td>q0, q3</td> <td>q1</td> </tr> <tr> <td>q3</td> <td>q2</td> <td>-</td> </tr> </tbody> </table>	Current State	Input 'a'	Input 'b'	→ q0	q0, q1	q0, q2	q1	--	q3	q2	q0, q3	q1	q3	q2	-	4	K		
Current State	Input 'a'	Input 'b'																		
→ q0	q0, q1	q0, q2																		
q1	--	q3																		
q2	q0, q3	q1																		
q3	q2	-																		
Q-3	Find the re for the following automata:	3	A P																	
Q-4	For the following PDA find the sequence of moves for input: aaabbb $\delta(q_0, a, Z) = \{ (q_1, AZ) \}$ $\delta(q_1, a, A) = \{ (q_1, AA) \}$ $\delta(q_1, b, A) = \{ (q_2, \epsilon) \}$ $\delta(q_2, b, A) = \{ (q_2, \epsilon) \}$ $\delta(q_2, \epsilon, Z) = \{ (q_3, Z) \}$	4	K																	
Q-5	Construct a PDA which accepts strings made of a's and b's and it should be of type $a^n b^n$, where $n \geq 1$.	4	A P																	
Q-6	Derive the string "aaabbabbba" for the following grammar: $S \rightarrow aB \mid bA$; $S \rightarrow aS \mid bAA \mid a$; $B \rightarrow bS \mid aBB \mid b$	3	K																	
Q-7	Define Turing Machine and also explain Universal Turing Machine	1	U																	
Q-8	Design CFG for $a^n b^n$	2	E																	

Sessional 1 (B.Tech VI Sem CSE)
BCE-C625 Theory of Computation: Classical to Quantum Models

Time: 1HR

Max. Marks :20

Note: Attempt all questions.

Questions	Marks	Learning Outcome	Bloom's Taxonomy															
Q1 Define the term Finite Automata. Differentiate between NFA and DFA.	4	CO1	U															
Q2. Write regular expression for the language over the alphabet {0,1}. With no consecutive zeors.	4	CO1	U															
Q3. Design a DFA equivalent for the given NFA; where q3 is the final state. <table><tr><td>Current State</td><td>Input 'a'</td><td>Input 'b'</td></tr><tr><td>→ q0</td><td>q0, q1</td><td>q0 , q2</td></tr><tr><td>q1</td><td>--</td><td>q3</td></tr><tr><td>q2</td><td>q0, q3</td><td>q1</td></tr><tr><td>q3</td><td>q2</td><td>-</td></tr></table>	Current State	Input 'a'	Input 'b'	→ q0	q0, q1	q0 , q2	q1	--	q3	q2	q0, q3	q1	q3	q2	-	8	CO3	E
Current State	Input 'a'	Input 'b'																
→ q0	q0, q1	q0 , q2																
q1	--	q3																
q2	q0, q3	q1																
q3	q2	-																
Q4. Design a FA for (0*1*) +	4	CO2	K															

A- Analyze , U- Understand, AP - Apply, E-Evaluate, K-Knowledge

Sessional 2 (B.Tech VI Sem CSE)
BCE-C625 Theory of Computation: Classical to Quantum Models
Time: 1HR

Date : 22.04.2026

Max. Marks :20

Note: Attempt all questions.

Questions	Marks	Learning Outcome	Bloom's Taxonomy
Q1 Define PDA.	4	CO1	U
Q2. What is an ambiguous grammar.	4	CO1	U
Q3. For the given grammar do left most derivation for the string aaabbbb ; Also draw the parse tree. $S \rightarrow ab \mid ba \mid aB \mid bA$; $A \rightarrow a \mid bAA$; $B \rightarrow b \mid aBB$	8	CO3	E
Q4. Design a PDA made of a's and b's which accepts only palindrome.	4	CO2	K

A- Analyze , U- Understand, AP - Apply, E-Evaluate, K-Knowledge

PAPER CODE: BCE-C625
EVEN SEMESTER EXAMINATION (MAY-2026)
CLASS: B.TECH. SEMESTER: VI
PAPER TITLE: THEORY OF COMPUTATION:CLASSICAL
TO QUANTUM MODELS

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
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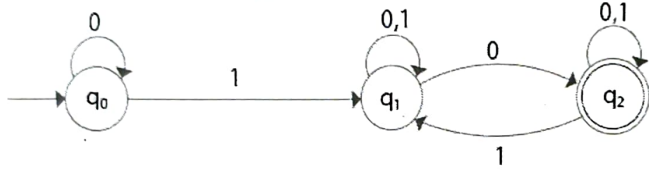
		CO	BL
Q-1	Construct a DFA over $\Sigma = \{0,1\}$ that accepts all strings ending with "01".	2	2
Q-2	Define Deterministic Finite Automaton (DFA). Write the formal 5-tuple representation of DFA.	3	3
Q-3	Differentiate between Moore Machine and Mealy Machine with suitable examples.	1	2
Q-4	Convert the following regular expression into NFA: (a+b)*ab	3	3
Q-5	Write a context free grammar (CFG) for the language: $L = \{a^n b^n n \geq 1\}$	1	2
Q-6	What is ambiguity in context free grammar? Give one example of ambiguous grammar.	2	1
Q-7	Construct a leftmost derivation for the string $a^5 a b b b b^5$ using the grammar: $S \rightarrow aSb \mid ab$	5	4
Q-8	Define Pushdown Automata (PDA). What is the role of stack memory in PDA?	2	2
Q-9	Explain the basic structure of a Turing Machine with the help of a diagram.	3	3

Q-10	Differentiate between decidable and undecidable problems with examples.	1	2
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SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 x 10 = 40 Marks)

		CO	BL
Q-1	Construct a DFA that accepts all binary strings containing an even number of 0's and an odd number of 1's. Also draw the transition table and explain the working of the automaton.	1	3
Q-2	Convert the following NFA into equivalent DFA. Show all steps. 	3	4
Q-3	Find the regular expression corresponding to the following language: "All strings over {a,b} that start with 'a' and end with 'bb'." Also design the equivalent finite automaton.	2	3
Q-4	Eliminate left recursion from the following grammar and construct the FIRST and FOLLOW sets: $A \rightarrow Aa A\beta \gamma$	3	4
Q-5	Convert the following CFG into Chomsky Normal Form (CNF): $S \rightarrow aAB b$ $A \rightarrow aA a$ $B \rightarrow bB b$	2	3
Q-6	Design a Pushdown Automaton for the language: $L = \{wcw^R \mid w \in \{a,b\}^*\}$ Explain each transition with stack operation.	3	4
Q-7	Explain Chomsky hierarchy of languages in detail.	1	6
Q-8	What is the Halting Problem? Prove that the Halting Problem is undecidable using diagonalization or contradiction method.	2	6